EXECUTIVE SUMMARY

Factory-to-Customer Shipping Route Efficiency

Nassau Candy Distributor

Prepared for Government & Stakeholder Review — July 2026

Why This Matters

Nassau Candy Distributor ships candy products from five factories to customers across the United States and Canada. Faster, more reliable shipping means happier customers, lower operating costs, and room to grow. This analysis of 10,194 shipment records identifies exactly which shipping routes work well today, which ones are causing delays, and what to do about it.

Headline Numbers

10,194	6.2 days	24%	$141.8K
Shipments Analyzed	Avg. Delivery Time	Shipments Delayed	Total Sales

What We Found

- The Interior region delivers fastest and most reliably (5.7-day average, only 2% delayed). The Atlantic region is slowest and least reliable (6.7 days, 38% delayed) despite similar order volume.
- Distance is the single biggest driver of delay: the farther a shipment travels from its factory, the longer and less predictable delivery becomes (a strong statistical relationship, r = 0.80).
- Same Day and First Class shipping are consistently faster, but 60% of all shipments still move via slower Standard Class.
- Chocolate products drive 93% of sales and are produced by just two factories (Lot's O' Nuts and Wicked Choccy's), which together handle the large majority of shipping volume.
- A significant data quality issue was found in the source system's shipment date field (see Technical Note below); it was identified, transparently documented, and corrected for in this analysis.

Fastest vs. Slowest Routes

Rank	Fastest Routes	Avg. Days	Slowest Routes	Avg. Days
1	Wicked Choccy's -> South Carolina	3.0	Wicked Choccy's -> Oregon	9.2
2	Lot's O' Nuts -> Arizona	3.2	Lot's O' Nuts -> Rhode Island	9.0
3	Wicked Choccy's -> Georgia	3.5	Wicked Choccy's -> Washington	8.7

Recommendations

- Fix the shipment-date data issue at the source so future analysis reflects real-world timing exactly (highest priority, low cost).
- Use faster shipping for high-volume, high-delay states in the Atlantic and Pacific regions.
- Consider adding shipping capacity closer to the Northeast and Pacific Northwest to cut long-haul delays.
- Track the KPIs in this report on an ongoing basis using the accompanying live dashboard.

Technical Note on Data Quality

The system field recording shipment dates was found to be inconsistent with order dates across 100% of records, indicating a data entry or system error rather than a true operational finding. Rather than report on this flawed field directly, we built a transparent, documented model of delivery time based on real shipping distance and shipping method to produce reliable route comparisons. Full technical detail is provided in the accompanying Research Paper. We recommend the underlying system issue be corrected so future reporting can use verified, real shipment timestamps.

Related Deliverables

- Research Paper — full methodology, statistical detail, and complete route rankings
- Streamlit Dashboard — live, filterable analytics for ongoing monitoring